

MANUEL NDEBELE

05 Sable Close, Borrowdale West

Harare, Zimbabwe

CELL: +263 773807928 / +26373947287

Email: manuza3993@gmail.com/graphmen.geo@gmail.com

INTERESTS AND CAREER GOALS:

ACCOMPLISHED DATA ANALYST/ENGINEER WITH EXTENSIVE SPATIAL DATA EXPERTISE

Experienced data analyst and engineer with nearly 10 years of expertise in working with spatial data. Demonstrated proficiency in developing GIS databases, spatial data capture, visualization, analysis, and storage. Skilled in geodatabase design and management, GIS algorithms and modeling, remote sensing, and image processing, including hyperspectral remote sensing for earth observation. Adept at leveraging advanced data analysis, business intelligence, and data mining techniques to extract valuable insights from complex spatial and non-spatial datasets. Highly proficient in relational database design, SQL programming, and the use of powerful analytical tools like R, Python, MATLAB, Excel, Looker, and Snowflake. Proven ability to plan, model, analyze, test, and implement data-driven solutions to support business objectives. Experienced in systems development for GIS algorithms and modeling in geoscience infrastructure management, as well as GIS applications for telecommunications, infrastructure, and fleet management. Strong background in spatial statistics and spatial analysis for social and environmental impact assessment.

CORE COMPETENCIES

- Data Analysis and Visualization
- Business Intelligence and Data Mining
- Relational Database Design and SQL Programming
- Python, R, and MATLAB Programming
- Excel, Looker, and Snowflake Proficiency
- Geodatabase Design and Management
- GIS Algorithms and Modeling
- Remote Sensing and Image Processing
- Spatial Statistics and Spatial Analysis
- GIS Solution Development and Implementation

TECHNICAL PROFICIENCIES / COMPUTER SKILLS:

- **Programming:** Python, R, SQL, Bash scripting
- **Data Analytics & Visualization:** Tableau, Power BI, Matplotlib, Plotly, Seaborn, Bokeh
- **Data Manipulation & Cleaning:** Pandas, NumPy, SciPy, Dplyr, data.table
- **Machine Learning:** Scikit-learn, TensorFlow, Keras, LightGBM, XGBoost
- **Databases:** SQL Server, PostgreSQL, MySQL, SQLite
- **GIS Tools:** ArcGIS, QGIS, Geospatial data processing
- **Cloud & Big Data:** AWS (S3, Athena, Glue, EMR), Google Cloud, Hadoop, Spark
- **Reporting & Dashboarding:** Jira, Confluence, Slack, Trello
- **Productivity:** Microsoft Office (Excel, PowerPoint, Word), Google Suite

EMPLOYMENT HISTORY AND RELEVANT WORK EXPERIENCE

Ministry of Transport and Infrastructure (Jan 2010 - Dec 2010)

Data Management Specialist

- Responsible for updating and digitizing field survey data using Microsoft Access database software
- Implemented data entry protocols and quality assurance procedures to ensure accurate and consistent data capture
- Developed and maintained database structures to facilitate efficient data storage, retrieval, and management
- Administered backup and recovery processes to safeguard the integrity of the organization's transportation infrastructure data

Ministry of Primary and Secondary Education (2011 - 2013) Geography Teacher

- Taught high school geography courses, fostering students' understanding of spatial concepts and data analysis
- Designed and delivered interactive lessons that incorporated geospatial data visualization and analysis
- Organized and supervised extracurricular activities related to geography and spatial data exploration
- Gained experience in data-driven teaching and the application of geographic information systems (GIS) in an educational setting

Bulawayo City Council (August 2016 - July 2017) - GIS Technician Intern

- Maintained and updated the city's comprehensive GIS master database, ensuring spatial accuracy and data quality through COGO mapping techniques and attribute data encoding
- Researched and analyzed a variety of engineering and real estate records to update the GIS data, leveraging permit and property management systems
- Provided GIS data and analysis to support various city departments and external stakeholders like architects, engineers, and utility companies
- Operated GIS/CAD workstations to update the database with new data from projects, implemented computer plotting, and maintained backups

Water Print Investment (July 2018 - June 2019) - GIS Technician (GIS Data Manager)

- Performed geospatial data management, analysis, and documentation to support the Liquid Telecoms project
- Prepared detailed maps and visualizations of network data for internal users
- Ensured GIS and Cartograph database integrity through QA/QC, corrections, and updates using various data sources
- Streamlined GIS workflows and acted as a liaison between departments to facilitate database development and improvement
- Assisted in designing technical architecture, requirements, and statistical models for data analysis
- Developed scripts and automated processes for data extraction, manipulation, and reporting using R programming
- Participated in learning and applying new statistical analysis techniques and methods

Plumtree Town Council (July 2019 - August 2019) – (GIS Data Manager/Programmer)

- Developed land cover/land use maps through visual image interpretation and supervised classification
- Created existing and proposed maps for village-level land use planning (VLUPs)
- Geo-referenced and prepared mosaics from satellite imagery and topographic sheets

- Performed on-screen digitizing of features like roads, rivers, and settlements to update maps
- Developed intervention package maps at the village level

Centre for Innovation in Technology (September 2019 - September 2020) - GIS Data Manager/Specialist/Programmer

- Developed GIS mapping applications and geoprocessing tools to meet client needs
- Analyzed end-user issues and provided tailored solutions for GIS applications
- Created demonstrations and user guides to explain GIS application usage to customers
- Provided maintenance and support for GIS applications and mapping systems
- Assisted with quality assurance checks and recommended updates or new techniques as needed

**ZIMBABWE NATIONAL GEOSPATIAL AND SPACE AGENCY (January 2021 - Present)
Geographic Information Science and Earth Observation Scientist/Programmer**

- Developed interactive web-based mapping applications and custom geoprocessing tools to support advanced spatial data analysis and visualization
- Utilized a range of GIS technologies, programming languages, and frameworks (e.g., ArcGIS, Python, JavaScript) to build scalable and user-friendly spatial data applications
- Engineered custom spatial analysis algorithms and models to extract meaningful insights from complex geospatial datasets
- Collaborated with stakeholders to understand their spatial data requirements and translated them into actionable GIS-powered solutions
- Delivered engaging presentations and demonstrations to educate customers on the capabilities of the GIS applications and the insights derived from spatial data analysis
- Maintained and enhanced GIS applications according to changing business needs, ensuring seamless integration with evolving spatial data requirements
- Assisted in implementing robust data integrity and quality assurance processes to ensure the reliability and accuracy of spatial data-driven analytics
- Continuously researched and evaluated emerging spatial analysis techniques, tools, and technologies to recommend innovative solutions that enhance the organization's spatial data analysis capabilities
- Worked effectively in cross-functional teams to develop high-quality GIS solutions that leverage spatial statistics, modeling, and visualization

EDUCATION:

University of Zimbabwe (Harare, Zimbabwe 2014-2018)

- BSc Honors in Geographical Information Science and Earth

Lupane State University (Lupane, Zimbabwe 2020-2022)

- BSc Special Honors in Monitoring and Evaluation - 2.1 class

African Regional Centre for Space Science and Technology Education in English (ARCSSTE-E 2022-2023), Nigeria

- Postgraduate Diploma in Remote Sensing and Geographic Information Systems 1st class.

Zimbabwe Institute of Diplomacy 2021

- Diploma in Diplomacy, Protocol, Etiquette, and Professional